

HABEEBSEC ENTERPRISE

Enterprise Network Infrastructure & Security Lab

Professional Technical Project Report

Project Type	Enterprise network design, implementation, security and troubleshooting
Platform	Cisco Packet Tracer
Architecture	Router-on-a-Stick + Core/Access Switching + VLAN Segmentation
Services	DHCP, DNS, FTP, SSH
Security	ACLs, SSH management, port security, segmentation
Implementation Scope	10 representative client/guest endpoints + 4 servers
Design Capacity	43 employee workstations + 4 servers
Report Date	03 October 2026

Portfolio note: This report documents what was actually designed, implemented, tested and learned. It deliberately separates design capacity from the representative Packet Tracer implementation and does not publish device passwords.

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Each major numbered section begins on a new page. Evidence is embedded only where it materially strengthens the technical record.

1. Executive Summary

HabeebSec Enterprise is a representative enterprise network laboratory built in Cisco Packet Tracer to turn networking theory into a practical, security-conscious implementation. The project models a headquarters environment containing Management, IT, Staff, Server and Guest network segments, with centralized routing and controlled inter-VLAN communication.

The business design calls for support of 43 employee workstations and four servers. The Packet Tracer implementation intentionally uses a representative subset—two Management PCs, two IT PCs, five Staff PCs and one Guest PC—plus the four servers. This distinction is important: the project validates architecture and controls using a manageable test population rather than claiming that all 43 workstations were physically instantiated.

The implementation combines 802.1Q VLAN trunking, router-on-a-stick inter-VLAN routing, DHCP relay, centralized DHCP, internal DNS, FTP, SSH-based device administration, port security and extended ACLs. Security policy was validated with both successful and intentionally blocked traffic tests.

The troubleshooting phase deliberately introduced realistic failures, including an incorrect access-port VLAN assignment and removal of VLAN 20 from an IT switch trunk. Both were diagnosed using evidence rather than guesswork and restored without redesigning the network.

The final state demonstrates a functional segmented network with documented limitations. The topology has no ISP/Internet path, so Internet egress controls could be designed but not empirically validated. The server-originated ACL is retained as a documented design artifact but is not applied because a stateless inbound ACL on the server VLAN would interfere with legitimate return traffic.

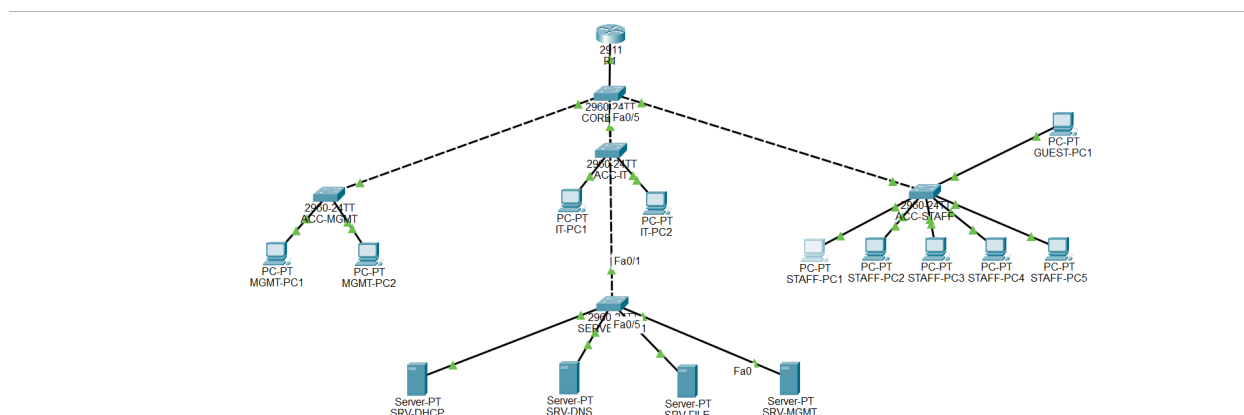


Figure 1. Final Packet Tracer topology showing the router, core switch, departmental access switches, server switch, representative endpoints and servers.

2. Requirements & Scope

2.1 Business and technical requirements

Requirement	Design response
Support 43 employee workstations	Use scalable /24 departmental VLANs; validate with representative endpoints in Packet Tracer.
Separate departments	Management, IT and Staff placed in distinct VLANs and IP networks.
Protect servers	Dedicated Server VLAN 40 with controlled access through router ACLs.
Provide guest connectivity	Dedicated Guest VLAN 50 with isolation from internal networks.
Centralize addressing	SRV-DHCP provides DHCP pools; R1 relays requests across VLANs.
Internal name resolution	SRV-DNS provides an A record for the File Server.
Secure administration	SSH enabled on R1 and switches; Telnet excluded from VTY input.
Limit unauthorized endpoints	Port security with sticky MAC and maximum one MAC per access port.
Demonstrate troubleshooting	Two deliberate faults introduced, investigated and restored.

2.2 Scope boundary

- **Included:** Layer 2 switching, VLAN segmentation, 802.1Q trunks, router-on-a-stick routing, DHCP, DNS, FTP, SSH, port security, ACLs, validation and troubleshooting.
- **Not implemented:** physical wireless infrastructure, WAN/ISP connectivity, Internet NAT/egress testing, enterprise wireless authentication, centralized logging/SIEM and production redundancy.
- **Evidence standard:** only completed configurations and observed tests are represented as implemented capabilities.

2.3 Design capacity versus implementation

Measure	Design requirement	Packet Tracer implementation
Employee workstations	43	9 employee PCs: 2 MGMT + 2 IT + 5 Staff
Guest endpoints	Scalable guest segment	1 Guest PC
Servers	4	4 servers
Total client/guest endpoints	43+ employees plus guests	10 representative endpoints

3. Architecture & Design

3.1 Logical architecture

The network follows a hierarchical enterprise-style layout. R1 provides Layer 3 gateway services for the VLANs. CORE-SW1 acts as the central Layer 2 aggregation point and carries the VLANs toward the departmental and server access switches. Departmental access switches provide endpoint access ports, while SERVER-SW1 isolates server-facing access ports.

3.2 Device roles

Device	Role	Key responsibility
R1	Router / policy enforcement point	Inter-VLAN routing, DHCP relay, SSH administration, ACL enforcement
CORE-SW1	Core/aggregation switch	VLAN transport and trunks to all downstream switches
ACC-MGMT	Management access switch	Management endpoint access, VLAN 10
ACC-IT	IT access switch	IT endpoint access, VLAN 20
ACC-STAFF	Staff/guest access switch	Staff VLAN 30 and Guest VLAN 50 endpoint access
SERVER-SW1	Server access switch	Server VLAN 40 and restricted management VLAN 10 transport

3.3 Why segmentation was used

Segmentation limits broadcast scope, improves fault isolation and provides a policy boundary. A compromised guest endpoint should not be placed in the same trust zone as administrative devices or servers. Similarly, the ability of IT to reach management resources is intentionally different from the ability of Staff or Guest users to do so.

3.4 Router-on-a-Stick

R1 uses one physical GigabitEthernet interface as an 802.1Q trunk. Subinterfaces G0/0.10, .20, .30, .40 and .50 terminate the five VLANs and provide their default gateways. This is appropriate for a Packet Tracer lab because it demonstrates the relationship between VLAN tags, routed interfaces and policy enforcement without requiring a multilayer switch.

```
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.40.10
ip access-group IT-IN in
```

```
R1>enable
R1#enable
R1#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       unassigned      YES unset    up              up
GigabitEthernet0/0.10    192.168.10.1    YES manual  up              up
GigabitEthernet0/0.20    192.168.20.1    YES manual  up              up
GigabitEthernet0/0.30    192.168.30.1    YES manual  up              up
GigabitEthernet0/0.40    192.168.40.1    YES manual  up              up
GigabitEthernet0/0.50    192.168.50.1    YES manual  up              up
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#
```

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Figure 2. R1 interface summary confirming all five VLAN gateway subinterfaces are up/up.

4. IP Addressing & VLAN Plan

4.1 VLAN and subnet plan

VLAN	Name	Subnet	Gateway	Purpose
10	Management	192.168.10.0/24	192.168.10.1	Administrative endpoints and switch management
20	IT	192.168.20.0/24	192.168.20.1	IT support/technical users
30	Staff	192.168.30.0/24	192.168.30.1	General employee endpoints
40	Servers	192.168.40.0/24	192.168.40.1	Infrastructure and application services
50	Guest	192.168.50.0/24	192.168.50.1	Untrusted/guest clients

4.2 Addressing convention

The design uses the VLAN ID as the third octet. The .1 address is consistently reserved for the router gateway. DHCP client allocations begin at .100 for the user VLANs, while servers use fixed addresses in the server subnet.

Host	Address	Method / role
SRV-DHCP	192.168.40.10	Static — DHCP service
SRV-DNS	192.168.40.11	Static — DNS service
SRV-FILE	192.168.40.12	Static — FTP/File service
SRV-MGMT	192.168.40.13	Static — management service host
MGMT-PC1	192.168.10.2	Static representative management endpoint
IT-PC2	192.168.20.101	DHCP representative IT endpoint
STAFF-PC1	192.168.30.100	Static representative staff endpoint
GUEST-PC1	192.168.50.100	DHCP representative guest endpoint

A /24 subnet provides 254 usable IPv4 addresses per VLAN, giving each segment substantial room beyond the representative Packet Tracer population and supporting the 43-workstation design requirement without changing the addressing model.

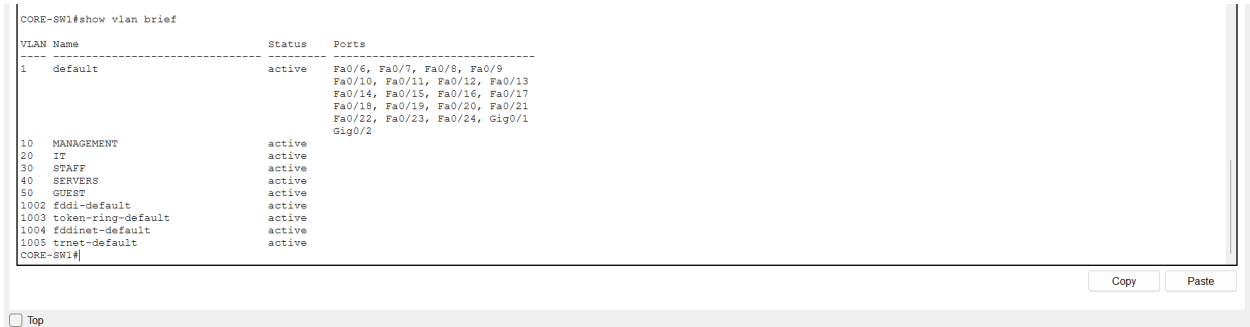


Figure 3. CORE-SW1 VLAN database showing VLANs 10, 20, 30, 40 and 50 active.

5. Infrastructure Configuration

5.1 Switching and trunks

CORE-SW1 uses five trunk links: one to R1 and one to each downstream switch. The access/server uplinks are also trunks. Endpoint-facing ports are configured as access ports and assigned to their appropriate VLAN. Trunk links are not protected with endpoint port-security because they are infrastructure links carrying multiple VLANs.

```
CORE-SW1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1
Fa0/5     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40,50
Fa0/2     1,10,20,30,40,50
Fa0/3     1,10,20,30,40,50
Fa0/4     1,10,20,30,40,50
Fa0/5     1,10,20,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40,50
Fa0/2     1,10,20,30,40,50
Fa0/3     1,10,20,30,40,50
Fa0/4     1,10,20,30,40,50
Fa0/5     1,10,20,30,40,50

CORE-SW1#
```

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Figure 4. CORE-SW1 trunk validation: five trunks are operational and VLANs 10/20/30/40/50 are active and forwarding.

5.2 Access-port allocation

Switch	Ports	VLAN
ACC-MGMT	Fa0/2–Fa0/3	10 Management
ACC-IT	Fa0/2–Fa0/3	20 IT
ACC-STAFF	Fa0/2–Fa0/6	30 Staff
ACC-STAFF	Fa0/7	50 Guest
SERVER-SW1	Fa0/2–Fa0/5	40 Servers

5.3 Server-switch trunk restriction

SERVER-SW1 intentionally carries only the VLANs needed by that switch: management VLAN 10 and server VLAN 40, in addition to the native/default VLAN behavior visible in Packet Tracer. It does not need user VLANs 20, 30 or guest VLAN 50. This is a least-privilege trunk design choice that reduces unnecessary VLAN propagation.

```
ACC-STAFF#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	
30	STAFF	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6
50	GUEST	active	Fa0/7
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ACC-STAFF#
```

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Figure 5. ACC-STAFF VLAN assignment showing Staff ports and the Guest endpoint isolated on VLAN 50.

6. Network Services

6.1 DHCP

SRV-DHCP provides separate pools for MGMT, IT, STAFF, SERVER and GUEST. R1 uses DHCP relay on client VLAN subinterfaces so broadcasts from remote VLANs can reach the centralized DHCP server. The server VLAN does not require a relay because SRV-DHCP is directly attached to that subnet.

Pool	Gateway	DNS	Start address	Mask
MGMT	192.168.10.1	192.168.40.11	192.168.10.100	/24
IT	192.168.20.1	192.168.40.11	192.168.20.100	/24
STAFF	192.168.30.1	192.168.40.11	192.168.30.100	/24
SERVER	192.168.40.1	192.168.40.11	192.168.40.100	/24
GUEST	192.168.50.1	192.168.40.11	192.168.50.100	/24

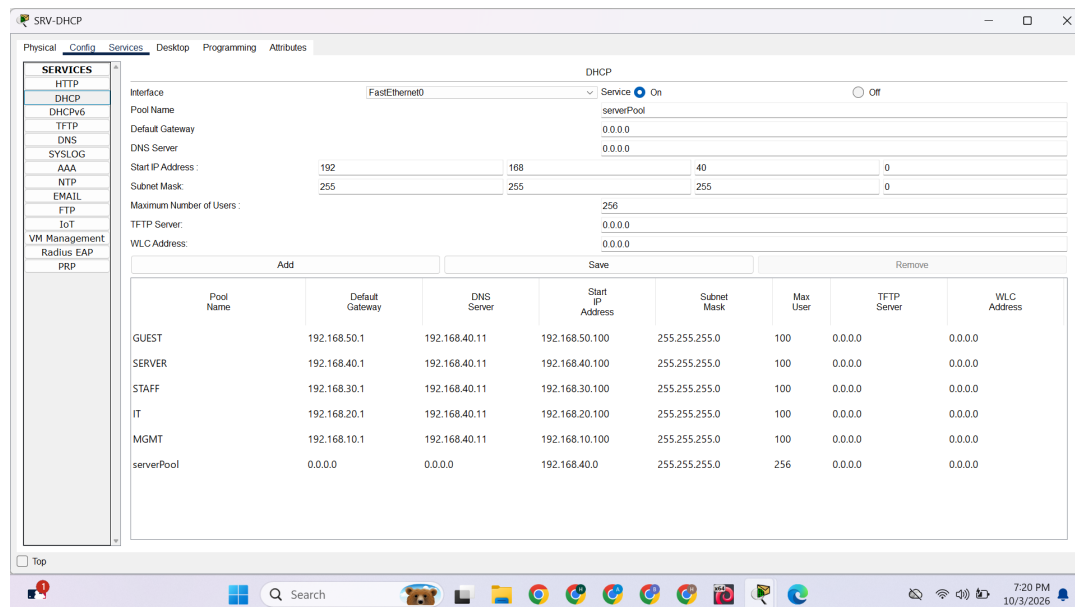


Figure 6. SRV-DHCP service showing the configured VLAN-specific address pools.

6.2 DNS

SRV-DNS hosts the internal A record files.habeebsec.local → 192.168.40.12. The record was validated from IT-PC2 with nslookup.

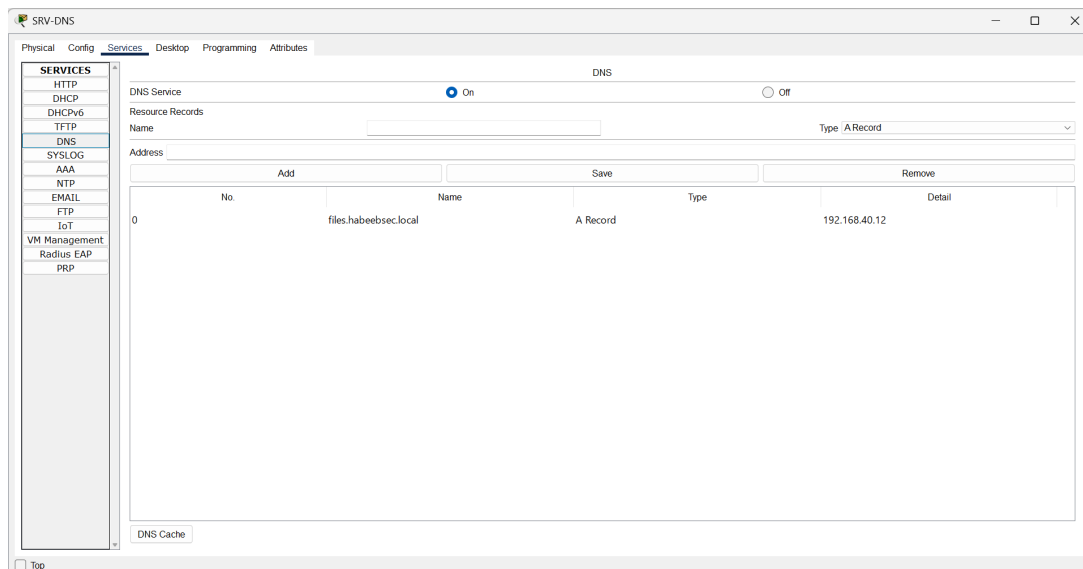


Figure 7. SRV-DNS showing the internal A record for files.habebsec.local.

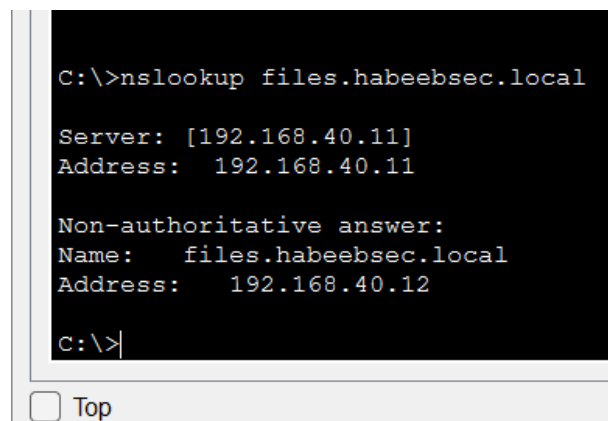


Figure 8. nslookup validation resolving files.habebsec.local to 192.168.40.12 through 192.168.40.11.

6.3 FTP

SRV-FILE provides FTP. A restricted user account was configured with Read/List permissions rather than write, delete or rename rights. IT and Staff access was tested successfully, including directory listing.

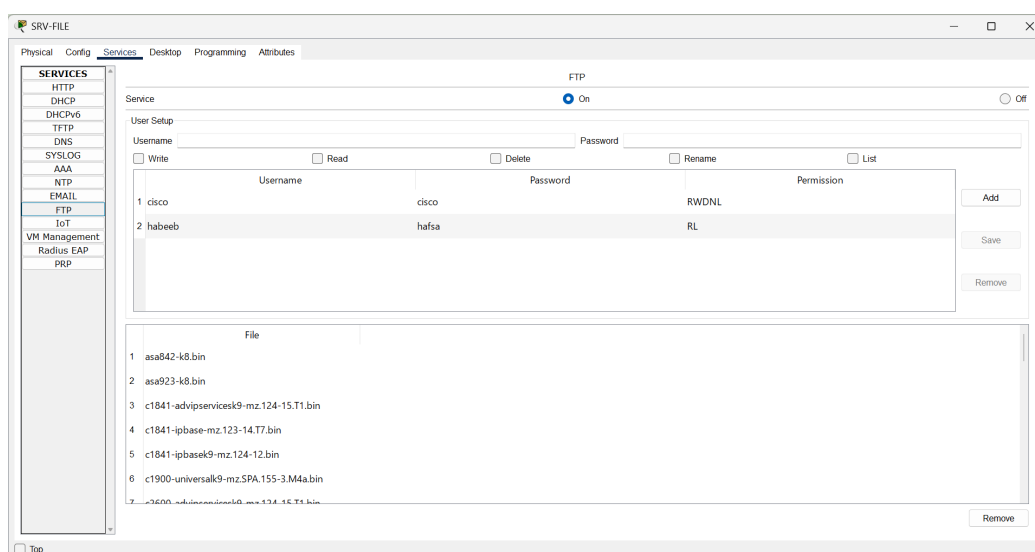
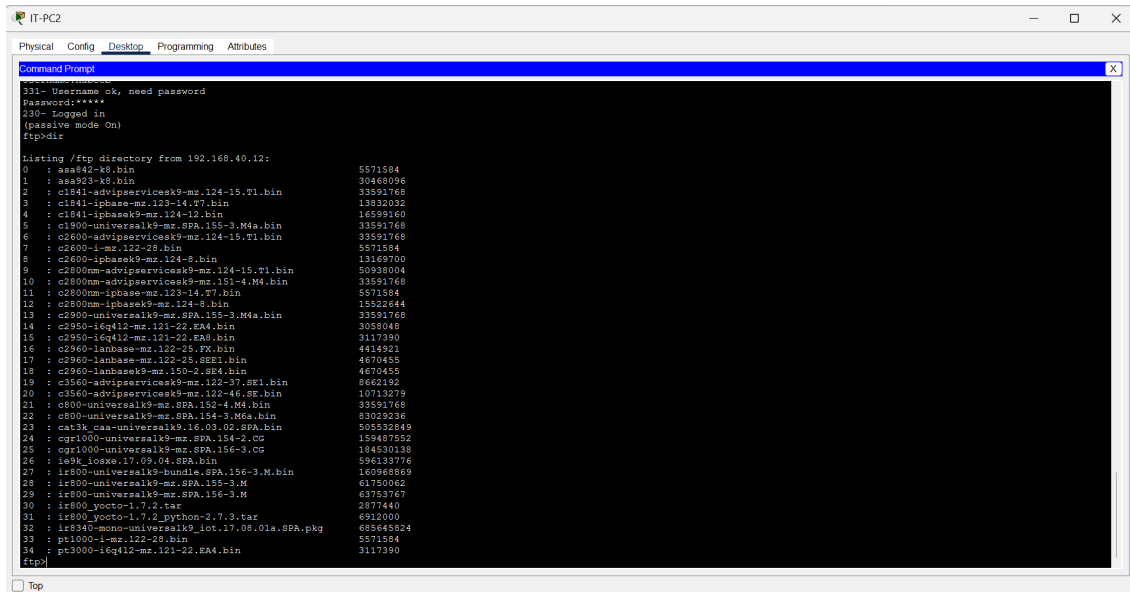


Figure 9. SRV-FILE FTP configuration showing the service enabled and a restricted user account.



```
IT-PC2
Physical Config Desktop Programming Attributes
Command Prompt
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.40.12:
0 : asa842-k8.bin 5571584
1 : asa823-k8.bin 30488086
2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3 : c1841-ibase-mz.123-14.T7.bin 13832032
4 : c1841-ibasek9-mz.124-12.bin 16399160
5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7 : c2600-1-mz.122-28.bin 5571584
8 : c2600-ibasek9-mz.124-8.bin 13169700
9 : c2800m-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800m-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800m-ibase-mz.123-14.T7.bin 5571584
12 : c2800m-ibasek9-mz.124-8.bin 15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2900-16q412-mz.121-22.EA4.bin 3058048
15 : c2950-16q412-mz.121-22.EA8.bin 3117390
16 : c2960-lanbase-mz.122-25.FX.bin 4414921
17 : c2960-lanbase-mz.122-25.SE1.bin 4670455
18 : c2960-lanbasek9-mz.150-2.S84.bin 4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE1.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532949
24 : egr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : egr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ie9k_iosxe.17.09.04.SPA.bin 596133776
27 : lr800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : lr800-universalk9-mz.SPA.155-3.M 61750062
29 : lr800-universalk9-mz.SPA.156-3.M 63753767
30 : lr800_yocto-1.7.2.tar 2877440
31 : lr800_yocto-1.7.2.pythom-2.7.3.tar 6832000
32 : lr8340-mono-universalk9_lot.17.08.01a.SPA.pkg 685645824
33 : pt1000-1-mz.122-28.bin 5571584
34 : pt3000-16q412-mz.121-22.EA4.bin 3117390
ftp>
```

Figure 10. IT-PC2 FTP directory listing from SRV-FILE, confirming authenticated access and data-channel operation.

7. Security Architecture

7.1 Security objectives

- Reduce trust between user populations through VLAN segmentation.
- Protect management interfaces from Staff and Guest networks.
- Permit only justified application traffic from user VLANs to servers.
- Use SSH instead of Telnet for device administration.
- Limit endpoint MAC addresses on access ports.
- Provide a default-deny posture in extended ACLs after required permits.

7.2 SSH management

R1 and the switches were configured for local user authentication, privilege level 15 administration, RSA keys and VTY lines restricted to SSH. An idle-session timeout was also configured. Device-specific administrative credentials were used; passwords are deliberately excluded from this public-facing report.

```
R1>
R1>enable
R1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R1#show running-config | section line vty
line vty 0 4
  exec-timeout 5 0
  login local
  transport input ssh
R1#
```

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Figure 11. R1 VTY configuration showing local authentication, SSH-only transport and a five-minute idle timeout.

```
C:\>ssh -l r1admin 192.168.10.1

Password:

R1#show privilege
Current privilege level is 15
R1#
```

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Figure 12. Representative SSH validation; privilege level 15 was confirmed after login.

7.3 Port security

Access ports use maximum one MAC address, sticky learning and shutdown violation mode. This reduces the risk of an unauthorized device being connected to a secured user-facing port. Trunks were intentionally excluded from endpoint port-security configuration.

```

ACC-STAFF#show port-security interface fa0/2
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000C.CF1B.99B4:30
Security Violation Count : 0

ACC-STAFF#

```

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Figure 13. Representative ACC-STAFF port-security validation: secure-up, maximum one MAC, one sticky MAC and zero violations.

7.4 ACL policy

Extended ACLs are applied inbound on the source VLAN subinterfaces of R1. This placement allows the policy to evaluate traffic close to its source before it traverses other internal segments. ACLs use first-match processing, so specific application permits appear before broader deny statements.

Source	Allowed examples	Blocked examples
Management	DNS, File Server, SSH	IT, Staff, Guest and broad server access
IT	DNS, File Server, SSH to devices, Management, Staff	Guest; broad Server access beyond permitted service rules
Staff	DNS, FTP/File Server	SSH to network devices, Management, IT, broad Server access
Guest	General permitted traffic outside protected internal ranges	Management, IT, Staff, Servers, SSH to R1
Servers	Not enforced with SERVER-IN in final state	Future policy requires state-aware refinement

```

R1#show access-lists
Extended IP access list STAFF-IN
 10 permit udp 192.168.30.0 0.0.0.255 host 192.168.40.11 eq domain (2 match(es))
 20 permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.11 eq domain
 30 permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.12 eq ftp (28 match(es))
 40 deny tcp 192.168.30.0 0.0.0.255 any eq 22 (48 match(es))
 50 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255 (4 match(es))
 60 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255 (12 match(es))
 90 permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.12 (5 match(es))
 100 deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255 (5 match(es))
 110 permit udp any eq bootpc any eq bootps
 120 deny ip any any
Extended IP access list IT-IN
 10 permit udp 192.168.20.0 0.0.0.255 host 192.168.40.11 eq domain (9 match(es))
 20 permit tcp 192.168.20.0 0.0.0.255 host 192.168.40.11 eq domain
 30 permit tcp 192.168.20.0 0.0.0.255 host 192.168.40.12 (73 match(es))
 40 permit tcp 192.168.20.0 0.0.0.255 any eq 22 (296 match(es))
 50 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 (4 match(es))
 60 permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 (8 match(es))
 70 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 (10 match(es))
 80 deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
 100 permit udp any eq bootpc any eq bootps (3 match(es))
 110 deny ip any any
Extended IP access list MGMT-IN
 10 permit udp 192.168.10.0 0.0.0.255 host 192.168.40.11 eq domain
 20 permit tcp 192.168.10.0 0.0.0.255 host 192.168.40.11 eq domain
 30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.40.12
 40 permit tcp 192.168.10.0 0.0.0.255 any eq 22 (40 match(es))
 50 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 60 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 70 deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
 80 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
 90 permit udp any eq bootpc any eq bootps
 100 deny ip any any
Extended IP access list SERVER-IN
 10 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
 30 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
 40 deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
 50 deny ip any any
Extended IP access list GUEST-IN

```

Figure 14. R1 ACL summary showing STAFF-IN, IT-IN, MGMT-IN and SERVER-IN rules with match counters.

```
R1#show access-lists GUEST-IN
Extended IP access list GUEST-IN
  deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255 (8 match(es))
  deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
  deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255 (8 match(es))
  deny ip 192.168.50.0 0.0.0.255 192.168.40.0 0.0.0.255 (5 match(es))
  deny tcp 192.168.50.0 0.0.0.255 host 192.168.50.1 eq 22 (35 match(es))
  permit ip 192.168.50.0 0.0.0.255 any (5 match(es))
  permit udp any eq bootpc any eq bootps
  deny ip any any
R1#
```

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Figure 15. GUEST-IN ACL showing explicit isolation from VLANs 10/20/30/40 and an explicit SSH deny to the R1 guest gateway.

8. Validation & Test Evidence

8.1 Validation matrix

Test	Source	Destination / action	Expected	Observed
DHCP	IT-PC2	DHCP lease	192.168.20.x + gateway + DNS	192.168.20.101 / .1 / .11
DNS	IT-PC2	files.habeebsec.local	192.168.40.12	Resolved successfully
FTP	IT-PC2	SRV-FILE	Login + directory listing	Successful
SSH	IT-PC2	R1	Permitted	Successful; privilege 15
SSH	Staff endpoint	R1	Blocked	Timed out / denied by ACL
Guest isolation	GUEST-PC1	MGMT/IT/Staff/Server	Blocked	Blocked
Port security	ACC-STAFF Fa0/2	Endpoint MAC	Secure-up, 1 MAC, 0 violations	Observed
Trunking	CORE-SW1	Fa0/1–Fa0/5	802.1Q trunking	Observed

8.2 Gateway validation

```
R1>enable
R1#enable
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    up          up
GigabitEthernet0/0.10    192.168.10.1    YES manual   up          up
GigabitEthernet0/0.20    192.168.20.1    YES manual   up          up
GigabitEthernet0/0.30    192.168.30.1    YES manual   up          up
GigabitEthernet0/0.40    192.168.40.1    YES manual   up          up
GigabitEthernet0/0.50    192.168.50.1    YES manual   up          up
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#
```

☐ Top

Figure 16. R1 final gateway state: G0/0.10 through G0/0.50 are all up/up with their planned gateway addresses.

8.3 Guest isolation evidence

```
GUEST-PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.20.101
Pinging 192.168.20.101 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.30.100
Pinging 192.168.30.100 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.30.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.40.12
Pinging 192.168.40.12 with 32 bytes of data:
Request timed out.
Request timed out.
```

Figure 17. Guest endpoint tests showing packet loss toward protected internal destinations, consistent with the GUEST-IN isolation policy.

9. Troubleshooting & Incident Log

Incident 1 — Incorrect access-port VLAN

Field	Record
Symptom	IT-PC2 lost expected IT connectivity.
Investigation	Checked endpoint addressing, then inspected ACC-IT with show vlan brief.
Evidence	Fa0/2 was found in VLAN 30 while Fa0/3 remained in VLAN 20.
Root cause	IT-PC2's access port had been assigned to the Staff VLAN.
Remediation	Restored ACC-IT Fa0/2 to VLAN 20.
Verification	IT connectivity and management-plane tests returned to expected behavior.

Incident 2 — VLAN 20 removed from IT trunk

Field	Record
Symptom	IT-PC2 could no longer reach R1/IT services.
Investigation	Checked ACC-IT Fa0/1 with show interfaces trunk.
Evidence	Allowed VLAN list excluded VLAN 20; active VLANs did not include VLAN 20.
Root cause	VLAN 20 was removed from the trunk's allowed list.
Remediation	Restored VLAN 20 with switchport trunk allowed vlan add 20.
Verification	IT connectivity returned without changing endpoint IP configuration.

Service recovery observation — DHCP

During evidence collection, IT-PC2 initially showed a DHCP failure and a 169.254.x.x autoconfiguration address. Investigation showed that the DHCP service/pool state in Packet Tracer was not operating as expected. The configured pools were re-saved/validated, R1's DHCP relay permit was adjusted to allow DHCP bootstrapping, and the endpoint subsequently received a valid 192.168.20.101 lease. This is documented as an implementation recovery event rather than hidden from the final report.

```
ip access-list extended IT-IN
permit udp any eq bootpc any eq bootps
deny ip any any
```

The engineering lesson is that a policy intended to protect a VLAN can accidentally block control-plane or bootstrapping traffic. Validation must include supporting protocols such as DHCP, not only business applications.

10. Security Findings & Improvements

10.1 Implemented controls

Control	Status	Evidence / effect
VLAN segmentation	Implemented	Five distinct security/trust zones
SSH-only VTY access	Implemented	Telnet excluded from VTY input
Device-specific admin accounts	Implemented	Credential reuse reduced
Port security	Implemented	1 MAC, sticky, shutdown violation
Source-side ACLs	Implemented	Role-specific inter-VLAN policy
Guest isolation	Implemented	Protected internal VLANs blocked
DHCP relay	Implemented	Centralized addressing across VLANs
Internal DNS	Implemented	Internal service name resolution

10.2 Limitations and next improvements

- **Internet egress:** no ISP/WAN exists in the current topology, so Guest/Staff Internet permission cannot be empirically tested.
- **Server-originated policy:** SERVER-IN was created but left unattached because broad stateless source-deny rules could disrupt legitimate return traffic. A production design would use stateful firewall controls or carefully scoped server-originated rules.
- **Management plane:** a dedicated management VRF or management firewall zone would strengthen isolation beyond the current VLAN/SVI model.
- **Redundancy:** the lab has single-router/single-core dependencies. A production design could add redundant gateways/switching and dynamic routing.
- **Monitoring:** Syslog, SNMP, NetFlow/telemetry and a SIEM integration would provide operational visibility.
- **Wireless:** Guest Wi-Fi is represented logically by VLAN 50, not by a simulated enterprise wireless controller/access-point deployment.

10.3 Design quality principle

The project intentionally avoids claiming production readiness. The portfolio value comes from showing that each implemented control has a reason, a configuration, a test and a known limitation.

11. Troubleshooting Methodology

11.1 Method used

- 1 **Define the symptom.** State exactly what is failing and what still works.
- 2 **Identify the path.** Map source endpoint → access port → trunk → gateway → destination.
- 3 **Check the nearest layer first.** Verify endpoint addressing and access VLAN before changing routing or ACLs.
- 4 **Compare expected versus observed state.** Use show vlan brief, show interfaces trunk, show ip interface brief, ACL counters and endpoint commands.
- 5 **Change one thing at a time.** Make the smallest corrective change that addresses the root cause.
- 6 **Retest the original symptom.** Do not stop after the configuration command succeeds.
- 7 **Record the lesson.** Capture the failure, evidence, root cause, remediation and verification.

11.2 Command evidence used

```
show vlan brief
show interfaces trunk
show ip interface brief
show access-lists
show running-config | section interface GigabitEthernet0/0.20
show port-security interface fa0/2
show ip ssh
show privilege
ipconfig
nslookup files.habeebsec.local
ping <destination>
```

11.3 Professional relevance

This methodology transfers to enterprise support, network operations, cloud networking and security operations. Cloud environments still contain interfaces, routes, security controls, name resolution, address allocation and policy boundaries; the physical device changes, but the diagnostic reasoning remains recognizable.

12. Final Architecture & Lessons Learned

12.1 Final architecture statement

The final lab is a segmented enterprise-style LAN in which R1 provides gateway and policy enforcement functions, CORE-SW1 aggregates VLAN trunks, access switches isolate departmental endpoints, SERVER-SW1 provides a restricted server segment, and centralized services provide DHCP, DNS and FTP. Management access is protected by SSH and access-port security.

12.2 Lessons learned

- A VLAN is a logical boundary, but the security policy becomes meaningful only when routing and ACLs enforce it.
- Trunk configuration is part of end-to-end connectivity; a correct access VLAN is useless if the VLAN is not propagated across the path.
- DHCP relay is necessary when a centralized DHCP server is located on another IP network.
- ACL order matters because Cisco ACL processing is first-match based.
- A broad stateless deny can break legitimate return traffic; direction and session behavior must be considered before applying an ACL.
- Packet Tracer can behave differently from production platforms, so evidence should describe observed simulator behavior rather than overgeneralize.
- Saving the Packet Tracer project and saving device startup configurations are separate responsibilities; both must be performed.

12.3 Final project status

Area	Status
Architecture	Complete
VLAN segmentation	Complete
Inter-VLAN routing	Complete
DHCP/DHCP relay	Complete and validated
DNS	Complete and validated
FTP	Complete and validated
SSH	Complete and validated
Port security	Complete and validated
ACL policy	Complete and validated
Troubleshooting labs	Complete — two distinct failures
Documentation	This report
Internet/WAN	Not implemented — future scope

13. Technical Design Rationale & Interview Defense

13.1 What problem was being solved?

The project solves the problem of designing a small enterprise LAN where departments have different trust levels and access requirements. The goal was not simply connectivity; it was controlled connectivity with evidence that the controls work.

13.2 Why VLANs?

VLANs provide logical separation without requiring a separate physical switch for every security zone. They also reduce broadcast scope and create clean policy boundaries for inter-VLAN routing.

13.3 Why router-on-a-stick?

It provides a clear demonstration of 802.1Q tagging, subinterfaces and centralized Layer 3 policy enforcement. It is appropriate for a lab of this scale, although a production environment with higher throughput requirements could use multilayer switching and redundant gateways.

13.4 Why DHCP relay?

DHCP discovery begins as a broadcast and normally does not cross a router. R1's ip helper-address converts that design into a routed request toward SRV-DHCP, allowing one centralized service to serve multiple VLANs.

13.5 Why ACLs on the source VLAN inbound?

Source-side filtering stops unauthorized traffic near its origin and makes the policy easier to reason about. It also avoids allowing unwanted traffic to traverse the network before being denied.

13.6 Why was SERVER-IN not left applied?

Because an ACL is stateless in this design. A broad inbound deny on traffic originating from the Server VLAN would also affect legitimate server responses to client-initiated sessions. The safer future design is a stateful firewall or precisely scoped server-originated rules.

13.7 Why only 10 endpoints if the requirement says 43?

The 43-workstation figure is the design capacity requirement. The Packet Tracer implementation uses representative endpoints to validate the architecture without creating an unnecessarily large simulator topology. The /24 subnet design preserves capacity for the intended population.

13.8 What would you improve next?

Add WAN/ISP connectivity, NAT and egress filtering; replace or supplement router ACLs with a stateful firewall; add redundancy; integrate logging/monitoring; expand wireless simulation; and automate configuration/documentation where practical.

14. GitHub Repository & Portfolio Packaging

The repository should present the project as an engineering artifact rather than as a collection of screenshots. The README should explain the business problem, architecture, implemented controls, validation and limitations before linking to deeper evidence.

14.1 Recommended repository structure

```
HabeebSec-Enterprise-Network-Security-Lab/  
■ ■ ■ ■ README.md  
■ ■ ■ ■ documentation/  
■ ■ ■ ■ technical-report.pdf  
■ ■ ■ ■ diagrams/  
■ ■ ■ ■ final-topology.png  
■ ■ ■ ■ configurations/  
■ ■ ■ ■ R1.txt  
■ ■ ■ ■ CORE-SW1.txt  
■ ■ ■ ■ ACC-MGMT.txt  
■ ■ ■ ■ ACC-IT.txt  
■ ■ ■ ■ ACC-STAFF.txt  
■ ■ ■ ■ SERVER-SW1.txt  
■ ■ ■ ■ evidence/  
■ ■ ■ ■ vlan/  
■ ■ ■ ■ services/  
■ ■ ■ ■ security/  
■ ■ ■ ■ troubleshooting/  
■ ■ ■ ■ project-files/  
■ ■ ■ ■ HabeebSec-Enterprise.pkt
```

14.2 Publication hygiene

- Do not publish real SSH passwords, FTP passwords or other reusable secrets.
- If configuration files contain credentials, replace them with clearly marked placeholders.
- Keep screenshots that demonstrate configuration, but crop unnecessary personal desktop information where possible.
- Do not claim that the lab implements WAN, wireless or production redundancy if it does not.
- Keep the Packet Tracer .pkt file with the project so reviewers can inspect the implementation.

14.3 Portfolio description

Enterprise Network Infrastructure & Security Lab — HabeebSec Enterprise. Designed and implemented a segmented enterprise LAN in Cisco Packet Tracer using VLANs, 802.1Q trunking, router-on-a-stick inter-VLAN routing, centralized DHCP/DHCP relay, internal DNS, FTP, SSH administration, port security and role-based extended ACLs. Validated connectivity and security policy through service tests and deliberate troubleshooting scenarios involving access-port VLAN assignment and trunk VLAN propagation.

15. Conclusion

HabeebSec Enterprise is complete as a practical enterprise-network security lab within the defined Packet Tracer scope. It demonstrates not only configuration ability but also design reasoning, security policy construction, validation, troubleshooting and documentation.

The strongest portfolio value is the engineering process: requirements were translated into VLANs and IP networks; routing and services were added; security controls were layered on top; failures were deliberately introduced and investigated; and the final state was tested against explicit expectations.

The project should now be treated as a baseline networking portfolio artifact. The next stage of the broader roadmap can build on this foundation by moving into Linux administration, automation and Git while retaining the same discipline: learn, practice, troubleshoot, build, document and defend.

Final statement: The report intentionally distinguishes designed capacity, implemented topology and validated behavior. That distinction is a core part of professional technical documentation.

Appendix A — Evidence Index

Evidence	Purpose	Report figure
Final topology	Architecture and implemented endpoint/server representation	Figure 1
R1 interface summary	Gateway/subinterface operational state	Figures 2 and 16
Core VLAN database	VLAN existence and naming	Figure 3
Core trunk state	802.1Q trunking and VLAN propagation	Figure 4
Staff/Guest VLAN ports	Endpoint segmentation	Figure 5
DHCP pools	Centralized address allocation	Figure 6
DNS record	Internal name resolution	Figure 7
nslookup	End-to-end DNS validation	Figure 8
FTP configuration	Application service and authorization	Figure 9
FTP directory listing	Authenticated service access	Figure 10
SSH VTY configuration	Secure device management	Figure 11
SSH validation	Administrative access test	Figure 12
Port security	Endpoint access-port hardening	Figure 13
ACL summary	Role-based traffic policy	Figure 14
Guest ACL	Guest isolation policy	Figure 15
Guest isolation test	Blocked internal access	Figure 17

Appendix B — Configuration Reference

R1 gateway addressing

```
G0/0.10 192.168.10.1/24 VLAN 10 DHCP relay ACL MGMT-IN
G0/0.20 192.168.20.1/24 VLAN 20 DHCP relay ACL IT-IN
G0/0.30 192.168.30.1/24 VLAN 30 DHCP relay ACL STAFF-IN
G0/0.40 192.168.40.1/24 VLAN 40 no relay SERVER-IN not applied
G0/0.50 192.168.50.1/24 VLAN 50 DHCP relay ACL GUEST-IN
```

Representative ACL logic

```
STAFF-IN
permit DNS to 192.168.40.11
permit FTP/File Server access to 192.168.40.12
deny SSH to any
deny access to Management
deny access to IT
deny broad Server access
permit DHCP bootstrapping
deny ip any any

IT-IN
permit DNS
permit File Server
permit SSH
permit Management
permit Staff
deny broad Server access
deny Guest
permit DHCP bootstrapping
deny ip any any

MGMT-IN
permit DNS
permit File Server
permit SSH
deny IT
deny Staff
deny broad Server access
deny Guest
permit DHCP bootstrapping
deny ip any any

GUEST-IN
deny Management
deny IT
deny Staff
deny Servers
deny SSH to 192.168.50.1
permit permitted traffic
permit DHCP bootstrapping
deny ip any any
```

Exact live configurations should be exported from the Packet Tracer devices and stored in the repository after removing or redacting credentials. This appendix is intentionally a design-level reference rather than a credential-bearing configuration dump.

Appendix C — Final Test Matrix

ID	Validation	Result
T01	All R1 VLAN subinterfaces up/up	PASS
T02	Core VLANs 10/20/30/40/50 active	PASS
T03	Core trunks operational	PASS
T04	IT-PC2 receives DHCP lease	PASS
T05	Guest-PC1 receives DHCP lease	PASS
T06	DNS resolves files.habeebsec.local	PASS
T07	IT FTP login + directory listing	PASS
T08	IT SSH to R1	PASS
T09	Staff SSH to R1 blocked	PASS
T10	Guest access to Management blocked	PASS
T11	Guest access to IT blocked	PASS
T12	Guest access to Staff blocked	PASS
T13	Guest access to Server VLAN blocked	PASS
T14	Guest SSH to R1 blocked	PASS
T15	Port security secure-up / zero violations	PASS
T16	Incorrect access VLAN troubleshooting	RESOLVED
T17	Missing trunk VLAN troubleshooting	RESOLVED

Testing reflects the representative implementation. It does not imply that all 43 design-capacity endpoints were individually tested.

Appendix D — Representative DHCP Client Evidence

This evidence confirms that the centralized DHCP service and R1 relay path successfully delivered a valid IT VLAN lease to the representative IT endpoint.

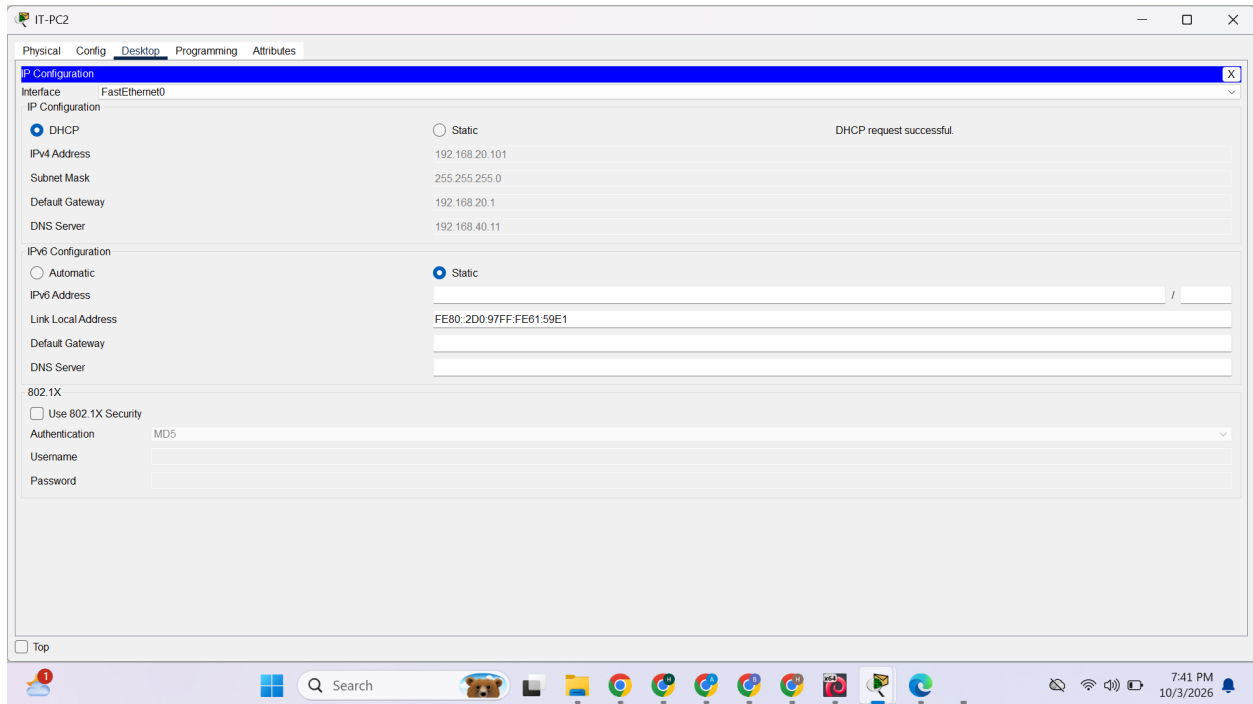


Figure D1. IT-PC2 received 192.168.20.101/24 with default gateway 192.168.20.1 and DNS server 192.168.40.11; Packet Tracer reports “DHCP request successful.”

This is representative validation, not a claim that every one of the 43 design-capacity workstations was individually leased and tested.